

01 · THE THESIS

Selection over speed.

In 2026 crypto markets, HFT firms own latency. After fees, most arbitrage signals are noise. Our edge is in deciding which cycles will actually execute as priced — not in detecting them faster.

POST-FEE FAILURE RATE

92.1%

of raw negative cycles fail the post-fee test

Binance pilot · BTC-ETH-USDT · 14 d

ROUND-TRIP TAKER FEE

COST

30 bps

Binance VIP-0 · 3 legs × 10 bps

RAW CYCLES SCANNED

VOLUME

≈ 2.4 k/s

Bellman-Ford on streaming book

POST-FEE SURVIVORS

SIGNAL

≈ 7.9 %

Forwarded to Confidence Gate

TEAM

Andrea Cammarano

Infrastructure & Cloud

*“Knowing when not to trade is the alpha.”***Giacomo Lanni**

Quant Strategy & Graph

Edoardo Praderio

ML Intelligence & Workflow

AI Associate (GPT-4o)

Code · Tests · Docs

02 · ARCHITECTURE

Triple-Gate execution pipeline.

Three independent rejection filters, AND-gated. A cycle fires only if every gate approves.

01 PROFIT GATE

Brain · math

Fee-aware filter

$ROI > \Sigma \text{ fees} + 5 \text{ bps buffer}$

REJECTS

92.1%

rejected before ML

LATENCY p50

<1 ms

02 CONFIDENCE GATE

Guardian · ML

Execution-success filter

$P(\text{fill within 5 bps}) > 0.85$

REJECTS

≈ 64%

of survivors rejected

LATENCY p50

3 ms

03 RISK GATE

Sentinel · risk

Inventory & drawdown filter

no cap breach · no halt

REJECTS

≈ 5%

rejected on inventory

LATENCY p50

<1 ms

EXECUTION PIPELINE



END-TO-END LATENCY (measured target)

MEDIAN

≈ 19 ms

p95

≈ 32 ms

p99

≈ 48 ms

03 · EVIDENCE & DELIVERY

Backtest, workflow, roadmap.

How we prove the gates work and how we deliver to spec by May 22.

BACKTEST

Walk-forward, fee-aware



FILL MODEL aggressive vs displayed depth + fee + 1-tick slippage

METRICS post-fee Sharpe · hit-rate · calibration · precision-at-threshold

MODEL · GUARDIAN

LightGBM primary + RF baseline

	RF	LGBM
Inference	6 ms	3 ms
Tabular AUC	0.78	0.82
Role	interpret.	production
Decision rule:	$\text{calibrated } \hat{p}(\text{isotonic}) > 0.85$	

ROADMAP



RUBRIC COVERAGE

ML ★ ADVANCED

REAL-TIME DATA

POSTGRES SQL DB

AGENTS.md + AI-PR

"The intelligence to decide when not to trade."